Jonathan Davidson

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Education

California State University, Los Angeles
M.S. in Mathematics, December 2025
Thesis: *Extremal Problems on Hamming Graphs* (PDF)
Advisor: Jason O'Neill

Stanford University
B.S. in Mathematics, June 2020

Research Interests

Combinatorial optimization; algebraic, geometric, and extremal graph theory; experimental and computational methods in combinatorics.

Papers and Manuscripts

Preprints

1. **J. Davidson.** *Coloring Monomial Cayley Graphs: The One-Parameter Case.* 2026. arXiv:2610.09474.
2. F. Ahmed, D. Asraf, D. Bonds, **J. Davidson**, Y. Jang, M. Krebs, A. Prakash, and E. Yak-De Padua. *Chromatic numbers of circulants with indispensable generators.* 2026. arXiv:2607.23841. Submitted.

Manuscripts in Preparation

- **J. Davidson**, J. O'Neill, and G. Ruiz-Navarro. *Optimal Monochromatic Rectangle Free Colorings of Grids.*
- **J. Davidson.** *Turán Densities in Hamming Graphs.*
- **J. Davidson**, G. Johnston, J. O'Neill, and G. Ruiz-Navarro. *Odd Representations of Hypergraphs.*

Research Experience

Current Research in Combinatorics

- Investigating monomial Cayley graphs as an algebraic approach to unit-distance graph coloring and the Hadwiger-Nelson problem.
- Studying extremal edge densities in Hamming graphs as alphabet size and dimension vary, connecting classical and hypercube extremal theory.
- Collaborating on monochromatic rectangle-free grid colorings and odd representations of hypergraphs.
- Using computational experimentation, SAT-based searches, and flag algebra methods to investigate constructions, conjectures, and extremal bounds.

Master's Research in Combinatorics

California State University, Los Angeles | January 2024 - December 2025

- Investigated extremal problems on Hamming and other product graphs under the direction of Jason O'Neill.
- Studied matrix methods for Abelian Cayley graphs with Mike Krebs.
- Explored the Union-Closed Conjecture, the Erdős Box Problem, Oddtown-type problems, and hypercube extremal questions.

Stanford University Research in Mathematics (SURIM)

Stanford University | Summer 2016

- Researched equilibrium capillary surfaces with Robert Finn.

Research Talks

Monomial Cayley Graphs: An Algebraic Approach to the Hadwiger-Nelson Problem

- Joint Mathematics Meetings (JMM), AMS Special Session on Algebraic Methods in Graph Theory and Applications | January 2027 | *Upcoming*

Multicolor Multipartite Ramsey Theory: A Combinatorial Design Approach

- SoCal Discrete Mathematics Symposium | University of California, Irvine | April 6, 2025 | Slides (PDF)
- Graduate Students Combinatorics Conference (GSCC) | University of Southern California | March 15, 2025 | Slides (PDF)

Monochromatic Rectangle Free Grids: A Design Theory Approach to the Erdős Box Problem

- USC Combinatorics Seminar | University of Southern California | November 18, 2024 | Slides (PDF)

Awards and Honors

NSS Outstanding Graduate Student Award
California State University, Los Angeles | November 2025

Phi Kappa Phi Honor Society
July 2025

Outstanding Presentation in Physical and Mathematical Sciences
California State University, Los Angeles | February 2025

Teaching Experience

SAT/AP Test Prep Instructor
Elite Prep, Arcadia, CA | December 2021 - Present

- Develop and teach curriculum for SAT, AP Calculus, AP Physics, and AP Chemistry.

Teaching Associate
California State University, Los Angeles | August 2024 - December 2025

- Taught Math 1081 (Pre-Calculus) and Math 1085 (Discrete Mathematical Models).

Mathematics Tutor
Math Gym, California State University, Los Angeles | August 2024 - December 2025

- Tutored Pre-Calculus and Calculus to support first-year students.

Outreach and Expository Talks

PCC π Day
Pasadena Community College | March 2024, 2025, and 2026

Probability Games and π
PCC STEM Saturday, Pasadena Community College | April 27, 2024

Dice Games and the Central Limit Theorem
AVID STEM Outreach, California State University, Los Angeles | March 4, 2024

Additional Training

EXCILL V: Extremal Combinatorics at Illinois - Summer Schools
University of Illinois Urbana-Champaign | June 2026

- Participated in the Design Theory summer school on June 8 and the Flag Algebra summer school on June 12. Program

UCLA-CSU Summer Bridge Program
University of California, Los Angeles | Summer 2025

- Studied real analysis, linear algebra, Bayesian inference, algebraic geometry, and representation theory in a five-week intensive program.